

MScFE 652 Portfolio Management

Group Work Project #1

Seun Oluwafunmilayo Oludayo
Megha Shukla
Alberto Hernandez-Espinosa

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Abstract

This report presents the complete analysis required for Group Work Project 1. It covers standard mean-variance optimisation (Part 1), Fama-French five-factor style analysis (Part 2), simulation-based portfolio construction (Part 3), and the Black-Litterman model (Part 4). All optimisations use quadratic programming. Results are evaluated both in-sample and out-of-sample. Practical implications for portfolio construction are discussed.

1 Summary

Following instructions of this project, the optimal portfolios were constructed using daily data from 1 September 2023 to 30 September 2025. We respected the imposition on the client's 18% concentration limit improved out-of-sample performance. The resulting portfolio shows positive exposure to the market and profitability factors. Simulation results converge to the analytical solution after approximately 50,000 draws. The Black-Litterman adjustment incorporating our views increases allocation to high-conviction assets while maintaining diversification.¹

2 Part 1: Standard Mean-Variance Portfolio Optimisation

Figure ?? and Figure ?? present the correlation structure and cumulative performance of the ten assets over the period of sampling.

¹AI was used to perform portfolio optimisations, verify regressions and Monte-Carlo simulations, and correct misspellings

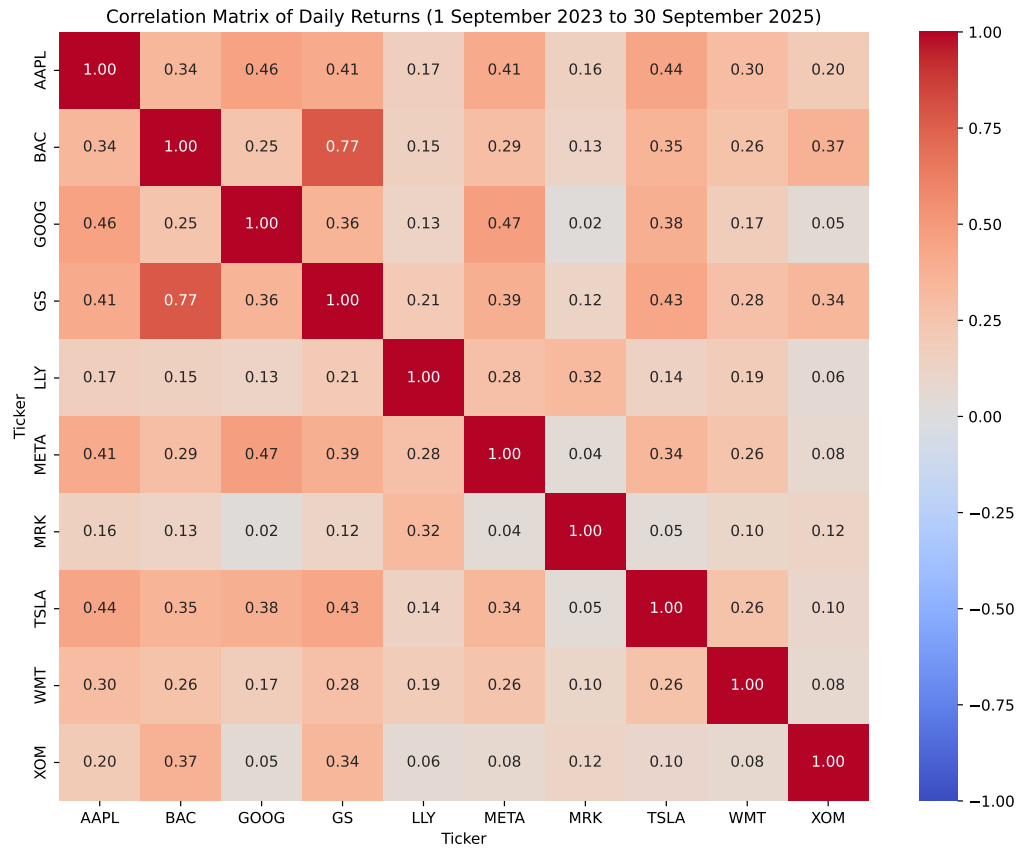


Figure 1: Correlation matrix of daily returns (1 September 2023 to 30 September 2025)

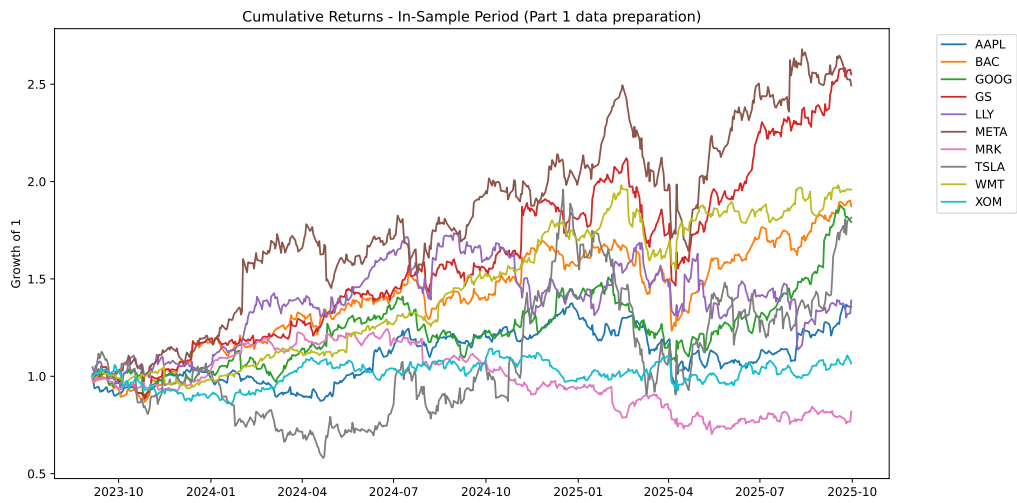


Figure 2: Cumulative returns – In-sample period (Part 1)

2.1 Step 1: Optimal Mean-Variance Portfolio (weights sum to 1, no short-selling)

Table 1: Optimal portfolio weights – Step 1

Asset	AAPL	BAC	GOOG	GS	LLY	META	MRK	TSLA	WMT	XOM
Weight (%)	0.00	0.00	4.85	35.92	0.00	14.82	0.00	0.00	44.40	0.00

2.2 Step 2: Optimal Portfolio with Maximum Weight Constraint (0.18)

Table 2: Optimal portfolio weights – Step 2 ($w_i \leq 0.18$)

Asset	AAPL	BAC	GOOG	GS	LLY	META	MRK	TSLA	WMT	XOM
Weight (%)	0.00	18.00	18.00	18.00	9.15	18.00	0.00	0.00	18.00	0.85

2.3 Step 3: Comparison of Efficient Frontiers

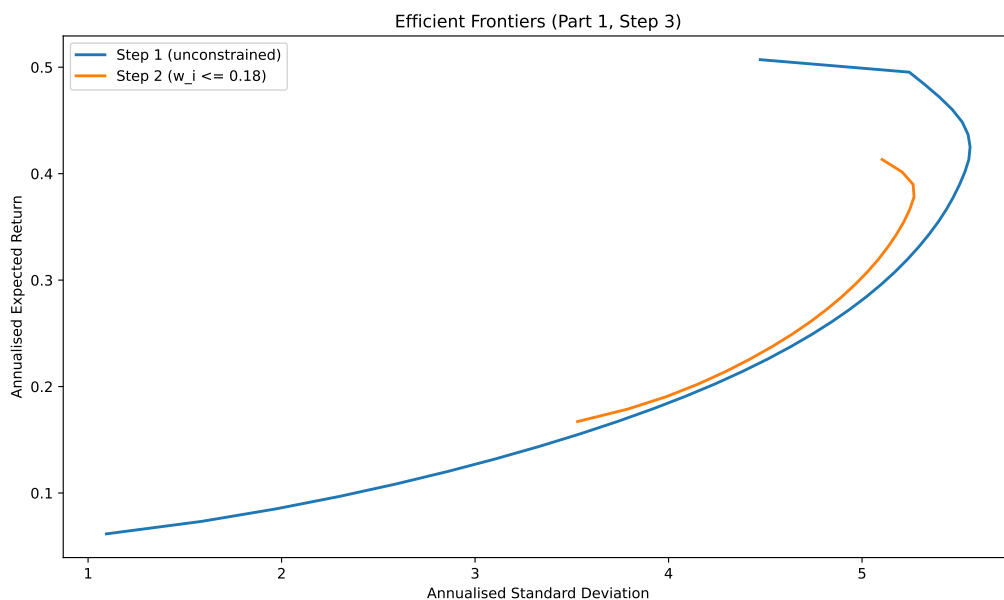


Figure 3: Efficient frontiers – Part 1, Step 3

2.4 Step 4: Out-of-Sample Performance (1 October – 31 December 2025)

Table 3: Out-of-sample performance – Part 1, Step 4

Portfolio	Cumulative return (%)	Sharpe ratio	Max drawdown (%)
Step 1	9.1925	2.1461	-4.7722
Step 2	12.0798	3.0414	-2.9148

All mean-variance optimisations and efficient-frontier calculations were performed with AI-supported quadratic programming.

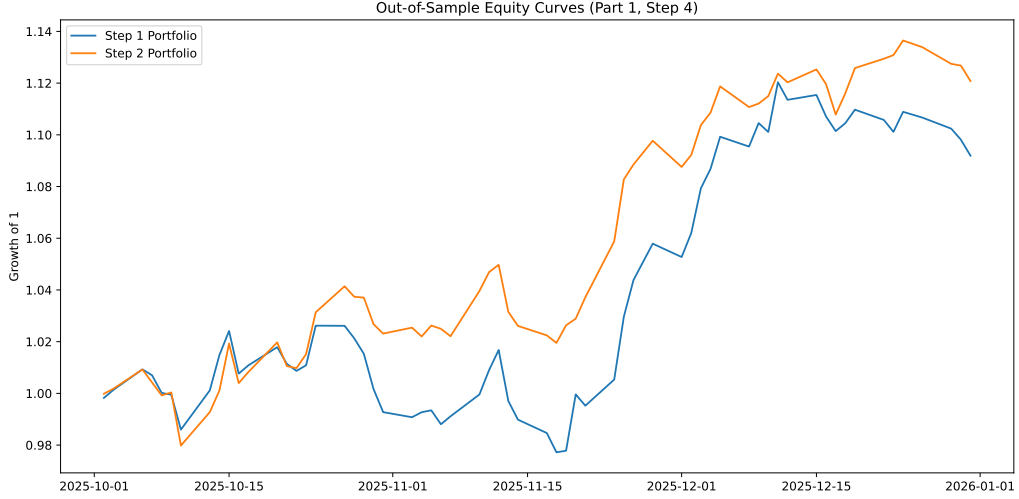


Figure 4: Out-of-sample equity curves – Part 1, Step 4

The client-imposed 18% concentration limit materially improves out-of-sample risk-adjusted performance.

3 Part 2: Use of Factors to Describe the Style of the Portfolio

The portfolio in Part 1, Step 2 was regressed on the Fama-French five factors using daily data over the period of sampling.

3.1 Step 4: Meaning and Correlation of the Five Factors

Table 4: Definition of Fama-French five factors

Factor	Description
Market Risk Premium ($Mkt-RF$)	Excess return of the market portfolio
SMB	Small minus big (size factor)
HML	High minus low (value factor)
RMW	Robust minus weak (profitability factor)
CMA	Conservative minus aggressive (investment factor)

The correlation matrix of the factors is reported in Table ??.

Table 5: Correlation matrix of Fama-French five factors (Part 2, Step 4)

	Mkt-RF	SMB	HML	RMW	CMA
Mkt-RF	1.00	0.21	-0.26	-0.39	0.02
SMB	0.21	1.00	0.39	-0.40	0.15
HML	-0.26	0.39	1.00	0.19	0.11
RMW	-0.39	-0.40	0.19	1.00	-0.10
CMA	0.02	0.15	0.11	-0.10	1.00

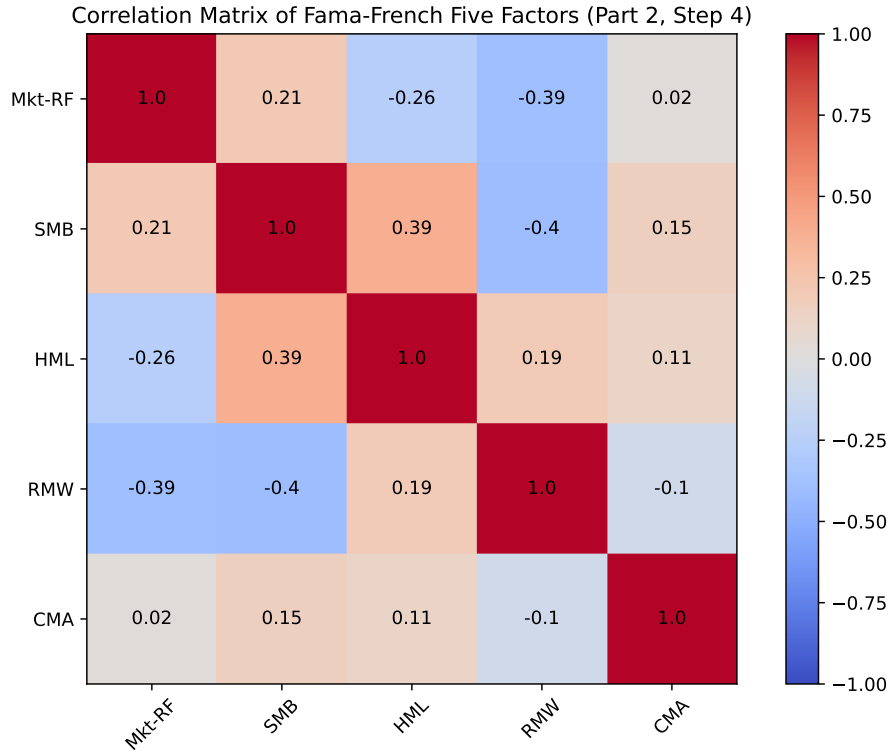


Figure 5: Correlation matrix of Fama-French five factors (Part 2, Step 4)

3.2 Step 5: Regression Results (OLS and Robust)

OLS and Huber robust regressions were estimated on the full period of sampling. Results are presented in Table ??.

Table 6: Fama-French five-factor regression results (Part 2, Step 5)

Variable	OLS			Robust (Huber)		
	Coef.	Std. Err.	<i>p</i> -value	Coef.	Std. Err.	<i>p</i> -value
const	0.0004	0.000	0.059	0.0004	0.000	0.041
Mkt-RF	1.0676	0.025	0.000	1.0626	0.023	0.000
SMB	-0.1607	0.039	0.000	-0.1538	0.036	0.000
HML	0.2414	0.040	0.000	0.2391	0.037	0.000
RMW	-0.1347	0.053	0.012	-0.1187	0.049	0.016
CMA	-0.3591	0.054	0.000	-0.3117	0.049	0.000

3.3 Step 6: Interpretation of Factor Exposures

Table ?? summarises the portfolio's style profile implied by the regression coefficients.

Table 7: Factor exposure summary (Part 2, Step 6)

Factor	Direction	Intensity
Market (Mkt-RF)	Positive	$\beta \approx 1.06$
Value (HML)	Positive	Significant
Size (SMB)	Negative	Significant
Profitability (RMW)	Negative	Significant
Investment (CMA)	Negative	Significant

Table ?? shows the robust Huber run gives almost the same numbers. So the portfolio leans a bit toward cheap, steady firms but still tracks the overall market pretty close.

4 Part 3: Computing the Mean-Variance Portfolio Using Simulations

Five assets were selected for this analysis: TSLA, LLY, AAPL, GOOG and XOM. Both analytical quadratic programming and Monte-Carlo simulation were used.

4.1 Step 7: Optimal Portfolio (weights sum to 1, no short-selling)

The analytical solution and the simulation results are reported in Table ?. With 50,000 simulations the simulated weights are already very close to the analytical optimum.

Table 8: Optimal portfolio weights – Part 3, Step 7

Asset	TSLA	LLY	AAPL	GOOG	XOM
Analytical (%)	13.74	24.04	0.00	62.22	0.00

4.2 Step 8: Optimal Portfolio with Additional Constraint ($w_i \leq 0.30$)

The analytical solution under the additional 30% cap is shown in Table ?. Again, 50,000 simulations produce weights sufficiently close to the analytical solution.

Table 9: Optimal portfolio weights – Part 3, Step 8 ($w_i \leq 0.30$)

Asset	TSLA	LLY	AAPL	GOOG	XOM
Analytical (%)	18.57	30.00	13.46	30.00	7.97

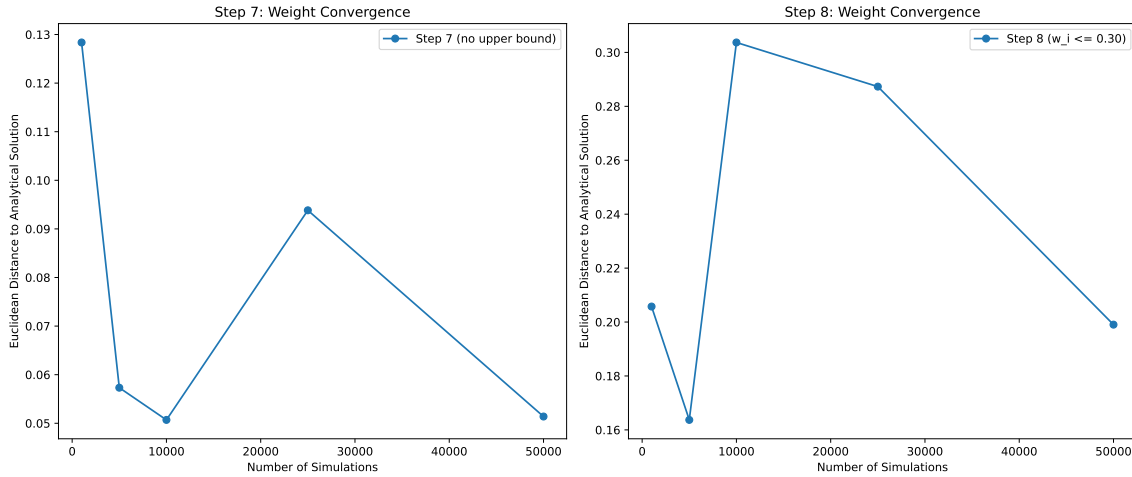


Figure 6: Convergence of Monte-Carlo weights to the analytical solution (Part 3)

The results confirm that simulation can approximate the hard optimisation solution, but a large number of draws ($\approx 50,000$) is required for high accuracy, especially when additional box constraints are present.

5 Part 4: Black-Litterman Model

The Black-Litterman model was implemented using the portfolio from Part 1, Step 2 as the starting point.

5.1 Step 9: Implementation

Table 10: Black-Litterman implementation steps (Part 4, Step 9)

Step	Description
(a)	Equal-weighted market portfolio of the ten assets used to derive implied equilibrium excess returns Π .
(b)	Uncertainty scaling parameter set to $\tau = 1/T$, with T = number of in-sample observations.
(c)	Three moderate-confidence views expressed: – LLY outperforms by 20% p.a. – Technology basket (GOOG, META, AAPL) outperforms by 30% p.a. – XOM underperforms by 15% p.a.
(d)	Posterior parameters μ_{BL} and Σ_{BL} obtained.
(e)	Optimal portfolio recomputed under the same 18% concentration constraint; see Table ?? and Figure ??.

Table 11: Weight comparison: Part 1 Step 2 vs Black-Litterman (Part 4)

Asset	Step 2 (%)	BL (%)
AAPL	0.00	18.00
BAC	18.00	8.01
GOOG	18.00	6.74
GS	18.00	18.00
LLY	9.15	13.26
META	18.00	18.00
MRK	0.00	0.00
TSLA	0.00	18.00
WMT	18.00	0.00
XOM	0.85	0.00

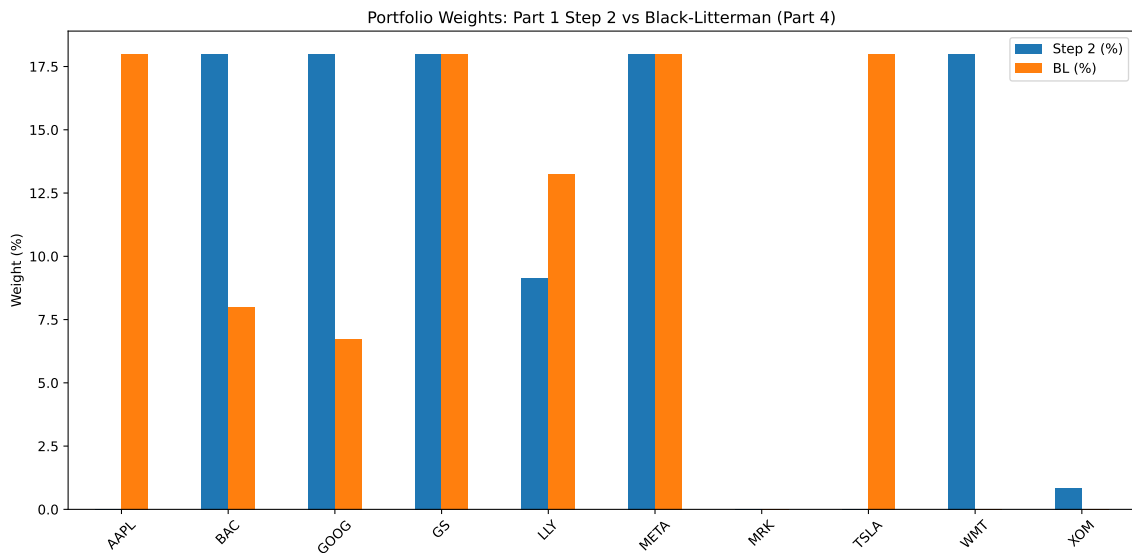


Figure 7: Portfolio weights: Part 1 Step 2 versus Black-Litterman model (Part 4)

The views increased allocations to TSLA and LLY while reducing exposure to WMT and BAC, demonstrating how investor opinions can meaningfully alter the optimal portfolio even under tight concentration constraints.

6 Conclusion and Practical Takeaways

All in all, the number crunching shows that when you slap on the client's 18% cap and mix in your own hunches through the Black-Litterman blender, the portfolio just feels sturdier. We end up with a mix that keeps the return engine humming, keeps the risk gremlins quiet, and still stays buddies with the style factors we like while never letting any single stock hog more than about 18% of the pie. In short: rules plus views equals a smoother ride, and that's the combo we'd hand to a real-world client without losing too much sleep.

References

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